



Entrepreneurship

CSC380

PROJECT NAME:

“CyberShield

Cybersecurity Awareness Training Platform”

PROJECT CARRIED OUT

BY

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1. Executive Summary

CyberShield is an interactive, web-based cybersecurity awareness training platform developed as an entrepreneurship project at the Institute of Business Management (IoBM). The platform is designed to educate students, employees, and general users about real-world cyber threats through structured lessons, interactive quizzes, phishing simulations, and gamified progress tracking.

In an era where cyberattacks, phishing scams, and data breaches are at an all-time high, cybersecurity literacy has become a critical life skill. CyberShield addresses this gap by providing an engaging, accessible, and self-paced learning environment — entirely browser-based with no installation required.

The platform covers six core security domains: Phishing Awareness, Password Security, Social Engineering, Malware & Ransomware, Online Privacy, and Safe Browsing. Users progress through lessons, earn badges, complete quizzes, and test themselves against realistic email phishing simulations — making learning both effective and enjoyable.

2. Problem Statement

Cybercrime has become a global epidemic. According to industry reports, over 90% of successful data breaches begin with a phishing email, and the vast majority of victims are ordinary users — not organizations targeted by sophisticated hackers. Despite this, cybersecurity education remains largely inaccessible, jargon-heavy, and unengaging for the average person.

Key Problems Identified:

- Lack of cybersecurity awareness among students and young professionals in Pakistan
- Existing training tools are either too technical, too expensive, or too boring to be effective
- Phishing, weak passwords, and social engineering account for the majority of cyber incidents
- No widely available, free, gamified platform in Pakistan focused on practical cyber awareness
- Organizations waste significant resources due to preventable human-error-based breaches

3. Proposed Solution — CyberShield

CyberShield is a fully functional web application that makes cybersecurity training accessible, interactive, and measurable. Rather than passive reading, users actively learn by doing: they identify phishing emails in simulations, test password strength in real time, take randomized quizzes, and earn badges as they advance.

The platform operates entirely in the browser using local storage — no server or backend installation is required — making it instantly deployable in educational institutions, corporate environments, or as a self-study tool.

4. Features & Modules

Feature / Module	Description
User Authentication	Register/Login system with local storage. Admin account with elevated access to management dashboard.
Lessons (6 Modules)	Structured, accordion-style lessons covering: Phishing Awareness, Password Security, Social Engineering, Malware & Ransomware, Online Privacy, and Safe Browsing & Wi-Fi.
Interactive Quiz	Module-based quiz engine with a pool of 20 questions per topic. Each session randomly draws 10 questions with instant feedback and explanations.
Phishing Simulation	Six realistic email scenarios (3 phishing, 3 legitimate). Users classify each email and receive detailed clue-based feedback. Score tracked in real time.
Password Strength Checker	Live password analyzer computing entropy, character pool size, crack-time estimate (offline, 10B guesses/sec), and 7-point rule checklist.

Dashboard	Personalized progress dashboard showing score, badge count, module completion percentages, weak areas, and security risk level (Low/Medium/High).
Gamification & Badges	8 unlockable badges tied to lesson completion, quiz performance, and simulation excellence. Locked badges shown as motivation.
Admin Panel	Admin-only view displaying all registered users, aggregate statistics (total users, lessons completed, badges earned, total score), activity log, and user removal.
Phish Trap Page	Educational landing page that triggers when a user clicks a simulated phishing link, explaining exactly what happened and what to do next.

5. Lesson Content Overview

Each lesson module follows a consistent pedagogical structure designed to maximize knowledge retention:

- Introduction
What is this threat and why does it matter?
- What It Means
A plain-language analogy explaining the concept
- Red Flags / Best Practices
Practical, actionable tips (8 per module)
- Real-World Example
A documented or realistic incident illustrating the threat
- How to Protect Yourself
Specific, step-by-step protective measures

Module 1	Phishing Awareness Recognize fake emails, spoofed sites, and spear phishing
Module 2	Password Security Strong password creation, 2FA, password managers, breach response

Module 3	Social Engineering Pretexting, baiting, tailgating, vishing, authority manipulation
Module 4	Malware & Ransomware Virus types, ransomware lifecycle, infection vectors, removal
Module 5	Online Privacy Data harvesting, browser tracking, VPN usage, privacy settings
Module 6	Safe Browsing & Wi-Fi HTTPS, public Wi-Fi risks, router security, browser hygiene

6. Technical Architecture

CyberShield is built entirely with front-end web technologies, making it lightweight, portable, and instantly deployable without any server infrastructure.

Technology Stack

HTML5	Semantic markup for all 9 pages (index, login, dashboard, lessons, quiz, simulation, password-checker, admin, phish-trap)
CSS3	Single unified stylesheet (style.css) with a dark-themed design system using CSS custom properties, flexbox, and grid layouts
Vanilla JavaScript	7 modular JS files: auth.js, dashboard.js, lessons.js, quiz.js, simulation.js, password-checker.js, admin.js
localStorage API	All user data (credentials, progress, scores, badges) persisted in browser localStorage — zero backend required
VS Code + Chrome	Development environment with launch.json configured for local server debugging on port 8080

File Structure

- index.html — Landing / home page
- login.html — Authentication (register & login)
- dashboard.html + dashboard.js — User progress dashboard
- lessons.html + lessons.js — 6-module lesson content
- quiz.html + quiz.js — Randomized module quizzes (100 questions total)
- simulation.html + simulation.js — Phishing email simulation
- password-checker.html + password-checker.js — Live password analyzer
- admin.html + admin.js — Admin management panel
- phish-trap.html — Phishing trap educational page
- css/style.css — Global design system
- auth.js — Shared authentication utilities

7. Quiz Engine

The quiz system is one of the most technically sophisticated features of CyberShield. It contains a bank of 120 questions across 6 modules (20 per module), and randomly selects 10 per session to ensure variety on repeat attempts.

Total Questions	120 questions (20 per module × 6 modules)
Per Session	10 randomly selected questions per quiz attempt
Modules	Phishing, Passwords, Social Engineering, Malware, Privacy, Safe Browsing
Question Format	Multiple choice (4 options) with one correct answer
Feedback	Immediate per-question feedback with detailed explanation
Scoring	Tracked and saved to user profile; contributes to risk level assessment
Results	Module-by-module accuracy breakdown displayed on dashboard
Repeat Friendly	Different question set each time — promotes deeper learning

8. Phishing Simulation Engine

The phishing simulation presents users with six realistic email cards — three genuine and three phishing attempts — and asks them to classify each one. After each decision, the system reveals whether they were correct and lists the specific clues that identified the email as legitimate or malicious.

Simulated Emails Included:

- PayPal Security Alert (PHISHING) — Lookalike domain "paypa1-alerts.com", urgency tactic, generic greeting
- GitHub Email Verification (SAFE) — Official domain, username-personalized, no urgency
- IT Support Password Reset (PHISHING) — Requests credentials via email reply, external domain, extreme urgency
- Google New Sign-in Notification (SAFE) — Official domain, specific location/time, no action pressure
- Netflix Billing Failure (PHISHING) — Fake domain "netflix-payment-update.com", threat of permanent deletion
- LinkedIn Connection Requests (SAFE) — Official domain, routine notification, verifiable by direct login

Each incorrect classification costs points; correct answers award 50 points each. A perfect score of 300 unlocks the "Sim Expert" badge. Completing the simulation with 5+ correct answers also unlocks the badge.

9. Gamification & Motivation

CyberShield uses gamification mechanics proven to improve learning engagement and retention. Users are motivated by visible progress, unlockable rewards, and risk feedback.

Feature / Module	Description
8 Badges	Phish Spotter, Key Guardian, Mind Shield, Virus Hunter, Privacy Pro, Net Navigator, Sim Expert, Cyber Pro

Scoring System	Points awarded for quiz correct answers and simulation performance. Cumulative score visible on dashboard.
Progress Bars	Visual module completion bars (0–100%) shown per topic on the dashboard.
Risk Level Badge	Dashboard computes Low / Medium / High security risk based on quiz performance, with colour-coded indicator.
Weak Area Detection	System identifies and flags specific modules where the user scored below threshold.
Locked Badge Display	Unearned badges shown as greyed/locked — a motivating visual reminder of what is achievable.

10. Admin Panel

The Admin Panel provides a management interface accessible only to the administrator account. It allows monitoring of all registered users and platform-wide statistics.

- Aggregate Stats — Total users, total lessons completed, total badges earned, total cumulative score
- User Table — Name, username, score, badge count, lesson progress bar, status (Active/New)
- Search/Filter — Real-time search box filtering users by name or username
- User Removal — Admin can permanently remove any user account
- Activity Log — Recent activity feed showing lesson completions, badge earnings, and new registrations

Admin credentials are hardcoded for prototype purposes (username: admin, password: admin123). In a production deployment, these would be replaced with server-side authentication and hashed credentials.

11. Password Strength Checker

The Password Strength Checker is an educational tool embedded in the platform. As the user types a password, it performs real-time analysis and displays:

- Strength level — Very Weak / Weak / Fair / Moderate / Strong / Very Strong / Excellent
- Entropy calculation — Bits of entropy based on character pool size and password length
- Crack time estimate — Time to crack via offline brute-force at 10 billion guesses/second
- 7-Rule checklist — Minimum length, recommended length, uppercase, lowercase, numbers, symbols, no common patterns
- Colour-coded visual feedback — Red to green spectrum matching strength level

The checker computes the character pool (26 lowercase + 26 uppercase + 10 digits + 32 symbols = 94 max) and applies the formula: $\text{Combinations} = \text{pool}^{\text{length}}$; $\text{Crack time} = \text{Combinations} / 10,000,000,000$. Results are formatted from "Instantly" through to "Billions of years."

12. Current Limitations & Future Scope

Current Limitations

- Authentication uses localStorage — not suitable for production without a backend
- No real database — user data is not persisted across different devices or browsers
- Admin credentials are hardcoded — requires server-side security for real deployment
- No email/notification system — learner reminders or completion certificates not yet available
- Content is static — no CMS to add/update lessons without editing source code

Future Enhancements

- Backend integration (Node.js / Firebase) for persistent cloud-based user data
- Completion certificates with QR-code verification
- Mobile app version (React Native or PWA) for on-the-go learning
- AI-powered adaptive quizzes that focus on weak areas
- Additional modules: Ransomware Response, OSINT, Incident Reporting
- Urdu language support for wider reach across Pakistan
- LMS integration (SCORM-compliant) for institutional adoption

13. Conclusion

CyberShield represents a practical, immediately deployable solution to one of the most pressing digital challenges facing Pakistan and the broader developing world: cybersecurity illiteracy. By combining structured lessons, gamified quizzes, realistic simulations, and real-time tools, the platform transforms a traditionally dry subject into an engaging, measurable learning experience. The project demonstrates a viable entrepreneurship model a lightweight, scalable EdTech product with clear target markets, low development cost, and high social impact. With backend integration and content expansion, CyberShield has the potential to become a standard onboarding tool for universities and SMEs across the country.

Built with purpose. Designed for impact. Powered by awareness.